

Q8

The resultant gravitational field on the 1 kg mass is zero. Therefore, $\frac{GM_1M_2}{x^2} = \frac{GM_2M_3}{(15-x)^2}$

$$\frac{M_3}{M_1} = \frac{(15-x)^2}{x^2} \Rightarrow \sqrt{\frac{16}{9}} = \frac{15-x}{x}$$

$$\frac{4}{3} = \frac{15-x}{x} \text{ or } 4x = 45 - 3x$$

Hence $x = 6.4 \text{ m}$

Ans: C